SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5112

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5 **Total Marks:50**

Annexure A

CONCEPT MAPPING

Code of Conduct:

I Yamuna K/192511013 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Yamuna K

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Annexure A

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:
    a. Show how each attack leads to specific threats and risks
    b. Map how security goals help mitigate them Provide one real-world example for each link

2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
    a. Show differences and similarities
        i. Map how each affects the Data structure and Security level
        ii. Analyze which is more vulnerable to attacks and why

4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
    a. Show relationships between these concepts
    b. Analyze how each contributes to secure communication
    c. Include one real-world application for each

Concept Mapping

CO_1 Assessment Tool-2

Name : K. Yamuna

Reg No : 192511013

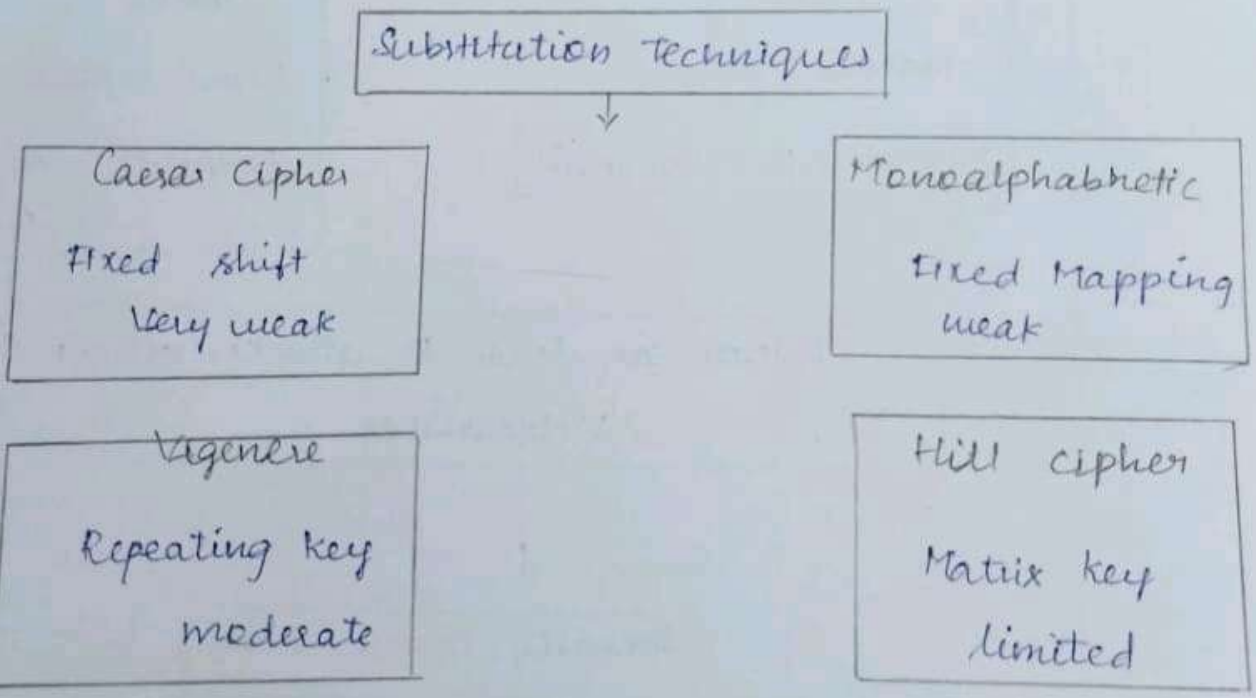
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Real-word example: phishing can expose passwords. HTTPS and encryption protect confidentiality, integrity checks detect changes, and backups support availability.

Conclusion: CIA security goals help reduce risks caused by cyberattacks.

2. Comparison of Substitution Ciphers.



Cipher	Complexity	Weakness
Caesas	low	Brute force
Monoalphabhetic	low	frequency analysis
Vigenere	Medium	Repeating key analysis
Hill	Medium-high	known-plaudert attack

Steganography and Information Protection

Concealing information and protecting digital content

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1. Data Hiding

Conceals secret information inside an ordinary file, such as an image, audio clip, or video.

Application: Hiding a confidential message inside an image sent online.

2. Digital Watermarking

Embeds identifying information in digital content to help establish ownership or verify authenticity.

Application: A photographer adds a watermark to protect and identify their images.

3. Survivability

The ability of hidden data or a watermark to remain detectable after changes such as compression, resizing, or format conversation.

Application: A robust copyright watermark remains detectable after an image is resized.

↓

Secure Communication

Concealment + Ownership identification + resistance to content changes.

Concept	Real-world example
Steganography	secret message hidden in an image
Data hiding	Confidential information embedded in media
Watermarking	photographer's logo on an image.
Survivability	watermark remains after image resizing

Conclusion

These techniques help conceal messages, protect ownership, and preserve embedded information. Encryption can provide additional security.

COMMON FEATURES

Both use a key or rule to transform plaintext into ciphertext, and both can be reversed by authorized decryption.

Substitution : Replaces letters, vulnerable to frequency analysis.

Transposition : Rearranges letters ; original letter frequencies remain unchanged.

Security: Both are vulnerable to cryptanalysis when used alone.

Conclusion: Substitution changes letter identity, while transposition changes letter order. Modern encryption provides better protection.

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4. Comept Map - Steganography Data Hiding, Watermarking and survivability.

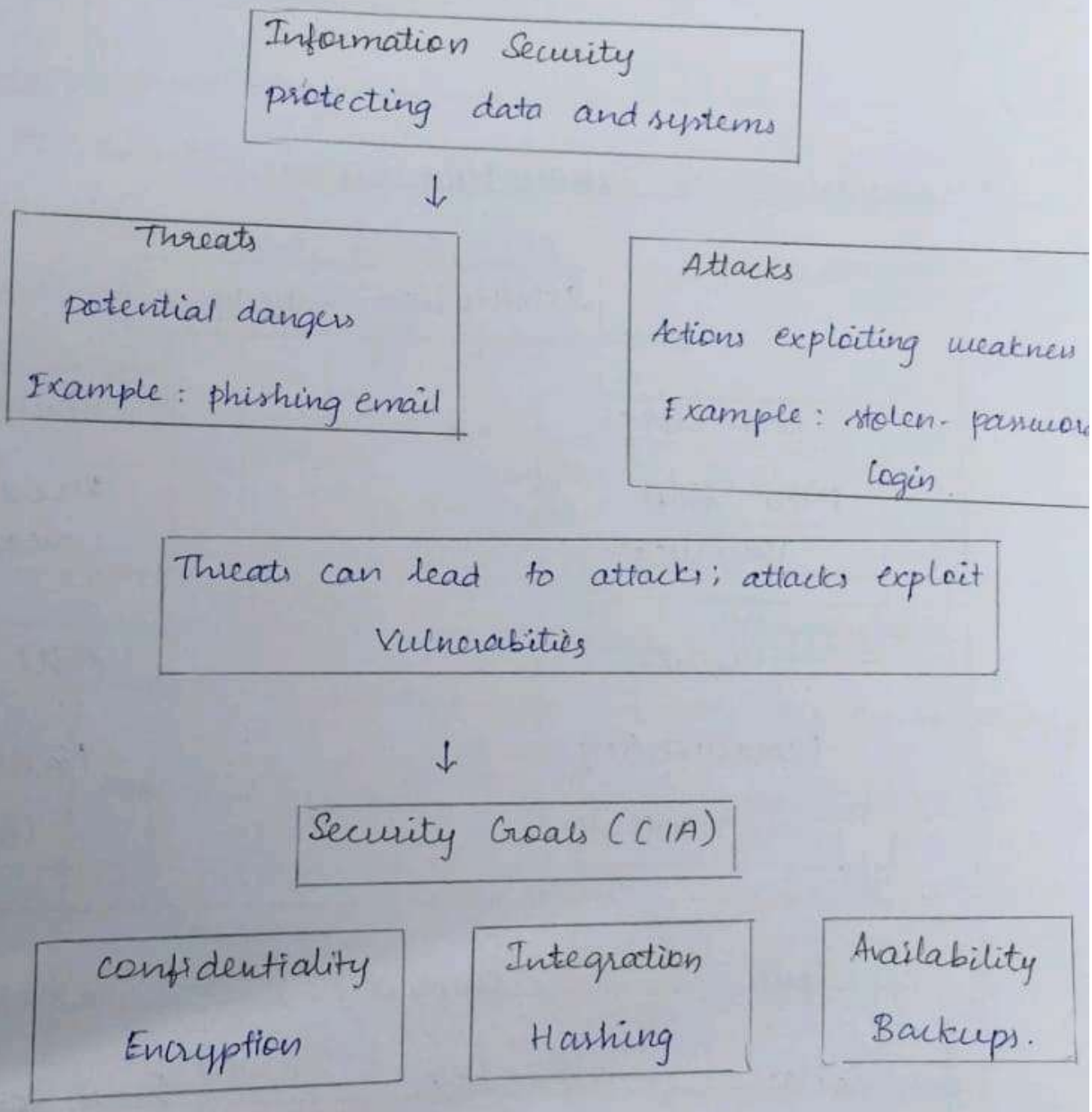
Aim

To connect steganography concepts and analyze how data hiding, watermarking, and survivability contribute to secure communication.

1. Security Attacks, Threats, Risks and Security Goals

Aim

To create a concept map connecting security attacks, threats, risks, and the security goals of Confidentiality, Integrity and Availability (CIA).



Conclusion: Vigenere and Hill can be stronger than simple substitution ciphers in some situation, but none is secure for modern communication

3. Substitution Vs Transposition

Aim

To identify the differences and similarities between substitution and transposition techniques, including their effects on data structure and security

Classical Encryption Schemes

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Substitution

Changes the symbols

Plaintext : HELLO

↓

Caesar (+3)

↓

KHOOR

letter identities change

positions generally remain

Transposition

Changes symbol positions

Plaintext : HELP

↓

Rearrangement

↓

HLEP

letter identities remain

positions change.

↓